Week 1 | CLO 1-A | File 2 / 7

1-1 Descriptive and Inferential Statistics

الإحصاء الوصفي والإحصاء الاستدلالي — البيانات والمتغير والمجتمع والعينة

1 · Why statistics, and what statistics is BOTH

Statistics is used in almost every field for decision-making: medicine (is a new drug effective?), education (what affects student performance?), business (predicting customer demand). Every one of these needs collected facts and observations, and those are **data**. AMANI

Statistics also reaches you through reports: the average height of Omani males is 171.70 cm and of females 158.44 cm; diabetes accounts for more than 15% of deaths among people aged 60 and over in Bahrain and Qatar; 51.7% of SQU students admitted in 2021–22 were in humanities colleges. AHMED

DEFINITION (EXAM WORDING)

Statistics is the science of conducting studies to **collect, organize, summarize, analyze**, and **draw conclusions** from data.

In your own research you must be able to design experiments; collect, organize, analyze and summarize data; make reliable predictions or forecasts; and communicate the results in your own words. AHMED

2 · The building blocks BOTH · CLO1-A

TERM	DEFINITION	EXAMPLE
Data البيانات	Raw facts, numbers, measurements or observations collected in a study; the values that variables can assume.	Heights of five students: 150, 165, 172, 180, 190 cm.
Data value (datum) قيمة البيانات	Each single value in the data set.	150 cm on its own.
Data set مجموعة البيانات	A structured collection of data values, usually arranged in a table.	The student-height table.
Variable المتغير	A measurable characteristic or attribute of an individual or object that can take different values.	GPA, attendance rate, study hours per week.
Random variable المتغير العشوائي	A variable whose values are determined by chance. AHMED	The number of heads in 3 coin tosses.

Remember AMANI: the *variable* is the category (GPA); the *data values* are what you record (3.7, 3.0).

variables →

one row = one individual

ID	College	Gender	GPA
10234	Science	Female	3.7
11234	Medical	Male	3.0
10771	Arts	Female	3.4

← a data value (datum): 3.0

the whole table = the data set

Dr. Amani's dataset activity answered on the table itself: each column heading is a variable, each cell is a data value (datum), and the whole table is the data set.

إجابة نشاط د. أماني على الجدول نفسه: كل عنوان عمود متغير (variable)، وكل خلية قيمة بيانات (datum)، والجدول كله مجموعة بيانات (data set).

3 · Population, sample, census, survey

BOTH · CLO1-A

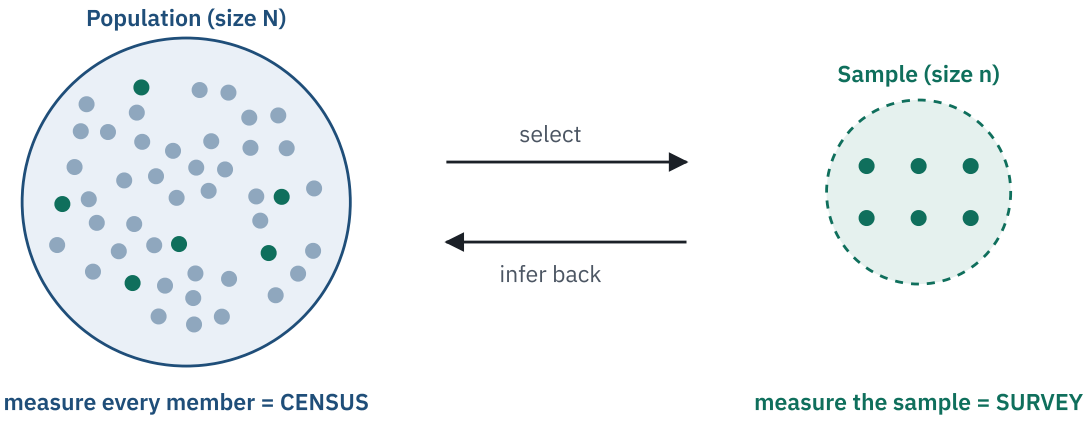
A **population** consists of **all** subjects or objects being studied, and must be clearly defined. Its size is `N`. Examples: all students at SQU; all cars a company produced in 2024; all registered STAT 2101 students; all SQU female students living with their families; all Lexus cars owned by SQU employees.

A **sample** is a group selected from the population, a **subset** of it. Its size is `n`. It must represent the population, so its subjects must be properly selected. Examples: 200 SQU students selected for a study; 150 cars tested for safety; 76 students randomly selected from the STAT 2101 list.

Collecting data from **every** member of the population is a **census**, such as Oman's General Census of Population, Housing and Establishments 2020 by the NCSI. Collecting from a sample is a **survey**.

Why samples? Expense, time, the size of the population, medical concerns and similar reasons usually make a census impossible.

A **biased sample** does not represent the population it came from. Example: choosing the students who arrive early to class. AHMED



A sample is a subset of the population. Measuring every member of the population is a census (e.g. Oman's 2020 census by the NCSI); measuring only a sample is a survey. Inferential statistics uses the sample to say something about the population.

العينة جزء من المجتمع. جمع البيانات من كل أفراد المجتمع يسمى حصرًا شاملًا (census) مثل تعداد عُمان 2020، وجمعها من عينة يسمى مسحًا (survey). الإحصاء الاستدلالي يستخدم العينة للحكم على المجتمع.

4 · The two branches

BOTH · CLO1-A

DESCRIPTIVE STATISTICS

Consists of the **collection, organization, summarization, and presentation** of data, from a sample or a population. Researchers describe a situation: collect data, organize and summarize it, and present it as charts, graphs or tables (Chapters 2 and 3).

INFERENTIAL STATISTICS

Consists of **generalizing from samples to populations**, performing estimations and hypothesis tests, determining relationships among variables, and making predictions. It is built on probability (Ch. 4–6), estimation and testing (Ch. 7–9), and relationships and prediction (Ch. 10). Its purpose is to make decisions and draw conclusions about the population.

Descriptive statistics

collect · organise · summarise · present

tables, frequency tables
graphs and charts (Ch. 2)
mean, SD, percentiles (Ch. 3)

"the average age in the 2020 census was ..."

Inferential statistics

sample → conclusion about population

probability (Ch. 4–6)
estimation, hypothesis tests (Ch. 7–9)
relationships, prediction (Ch. 10)

"...so flu cases next November will be ..."

The test is the verb. Describing what was observed (an average, a percentage, a chart of the collected data) is descriptive. Predicting, generalising to a population, or claiming a relationship goes beyond the data, so it is inferential.

المفتاح هو الفعل: إذا كانت الجملة تصف ما تم جمعه (متوسط، نسبة، رسم) فهي وصفية. وإذا كانت تتنبأ أو تعمّم على المجتمع أو تدّعي علاقة فهي استدلالية.

EXAMPLE 1-1 AHMED

Descriptive or inferential? (a) The average price of a 30-second Academy Awards ad was $1.90 million. (b) The Department of Economic and Social Affairs predicts Mexico City's population in 2030 will be 238,647,000. (c) A medical report: statins lower heart attacks, but some people are at slightly higher diabetes risk. (d) A Harris Poll of 2234 people: 55% said excessive complaining by adults was the most annoying social media habit.

(a) `Descriptive`: an average of observed data. (b) `Inferential`: a prediction. (c) `Inferential`: a general conclusion about everyone from studies. (d) `Descriptive`: a percentage of the 2234 people actually asked.

ACTIVITY · STATISTICS TYPE AMANI

(1) The Oman Meteorology Department records daily temperature variation in different cities and publishes a report on monthly weather patterns. (2) The Ministry of Tourism surveys visitor satisfaction at major attractions and uses the results to predict overall tourist satisfaction for the country.

(1) `Descriptive`: it reports what was recorded. (2) `Inferential`: a sample (visitors surveyed) is used to predict for the whole country.

Dr. Amani's diagram examples: the Oman Census 2020 average age and age frequency table are **descriptive**; the Ministry of Health surveying a sample of hospitals in November to predict next November's national flu cases is **inferential**.

الشرح بالعربي

الإحصاء **الوصفي** يجمع البيانات وينظمها ويلخصها ويعرضها كما هي: جداول ورسوم ومتوسطات. الإحصاء **الاستدلالي** يتجاوز البيانات: يعمّم من العينة على المجتمع، أو يتنبأ، أو يختبر فرضية، أو يدرس علاقة. أسرع اختبار في الامتحان: هل الجملة **تصف ما جُمع**؟ إذن وصفي. هل **تتنبأ أو تعمّم أو تدّعي سببًا**؟ إذن استدلالي. فخ شائع: «55% من 2234 شخصًا قالوا...» وصفي، لأنه يصف من سُئلوا فعلًا. ولو قيل «55% من كل البالغين» لصار استدلاليًا. وتذكّر: **المجتمع** هو الكل (حجمه N)، و**العينة** جزء منه (حجمها n). جمع البيانات من الكل = **census**، ومن عينة = **survey**.

Chapter 1 study set: 7 files. One merged source built from Dr. Amani Al-Rubkhi's Fall 2026 notes (course coordinator) and Dr. Ahmed Al-Hadhrami's slide pack, aligned to the SQU STAT 2101 course outline (Fall 2026). Textbook: Bluman, *Elementary Statistics*, 10th ed.

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